Cluster detection

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Plan

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Section 1

Motivation and Data

Motivation

- Interest of the topic: The question of spatial autocorrelation has received significant attention over time, with many techniques developed to better understand and detect it.
- This presentation is based on a research paper by (López, 2013) titled *"Exploring Scan Methods to Test Spatial Structure with an Application to Housing Prices in Madrid"*.
- We will present the method they propose to detect clusters, along with an application of this method to a different dataset. Through this example and the results we obtain, we will discuss the main advantages and limitations of this technique.

Data base (1/3)

- Before digging deep into the methodology and the example, let us present you the dataset we selected. Similarly to the one in the paper, we selected a dataset about house pricing not in madrid, but in boston.

- The dataset comes from **StatLib** and contains data on housing in Boston from 1970. It includes 506 observations and 21 variables. Since presenting all variables would not be very informative, we will focus on the main ones.

Data base (2/3)

Table: Variable description

Variable	Explanation	Type
CMEDV	median values of owner-occupied housing in USD	quantitative
CRIM	crime per capita	quantitative
ZN	proportions of residential land zoned for lots over 25000 sq. ft per town	quantitative
INDUS	proportions of non-retail business acres per town	quantitative
AGE	proportions of owner-occupied units built prior to 1940	quantitative
TAX	full-value property-tax rate per USD 10,000 per town	quantitative
PTRATIO	pupil-teacher ratios per town	quantitative

Data base (3/3)

Table: Descriptive statistics - Boston

Statistic	Mean	St. Dev.	Min	Max
CMEDV (1 000 USD)	22.53	9.18	5.00	50.00
CRIM	3.61	8.60	0.01	88.98
ZN (%)	11.36	23.32	0.00	100.00
INDUS(%)	11.14	6.86	0.46	27.74
AGE (%)	68.57	28.15	2.90	100.00
TAX (USD/10000)	408.24	168.54	187	711
PTRATIO	18.46	2.16	12.60	22.00

Section 2

Standard scan test

Standard scan test (1/2)

- Spatial and space-time scan tests are very popular in many fields and are still highly debated topics. Even definition-wise, there is no consensus yet. We will use the definition from (Kulldorff, 2009). The scan test is defined as follows:

  H_0: equality of the mean values of a variable x_i,with $i \in [1, n]$ across all location.

  H_1: There exists a location where the mean values is different.

Standard scan test (2/2)

- We can formally rewrite this hypothesis :
  $H_0 : \; x_i \sim N(\mu, \sigma) \quad \forall i \in [1, n]$

  $H_1 : \; x_i \sim N(\mu_Z, \sigma_Z) \quad (i \in Z) \quad \& \quad x_i \sim N(\lambda_Z, \sigma_Z) \quad (i \notin Z)$ with $\mu_Z \neq \lambda_Z$

- Even with this formalisation, it is not easy to visualise. Since understanding it is essential for the rest of this presentation, let's take a moment to explain it more simply.

Visualisation of the scan test (1/2)

- Let us consider a simple univariate case where we are working on a map M with the variable X
- We want to test for spatial autocorrelation following the definition of a scan test. We distinguish two main steps : construction & test
  - Construction : We draw a window Z on the map and compute the mean inside and outside the window. We do it a certain number of time while changing the **position, shape & the size** of Z each time (four for example). Once it is done we select the Z that maximises the difference in terms of means.
  - Test : We test if the difference between $\bar{X}_Z$ and $\bar{X}_{\not Z}$

Visualisation of the scan test (2/2)

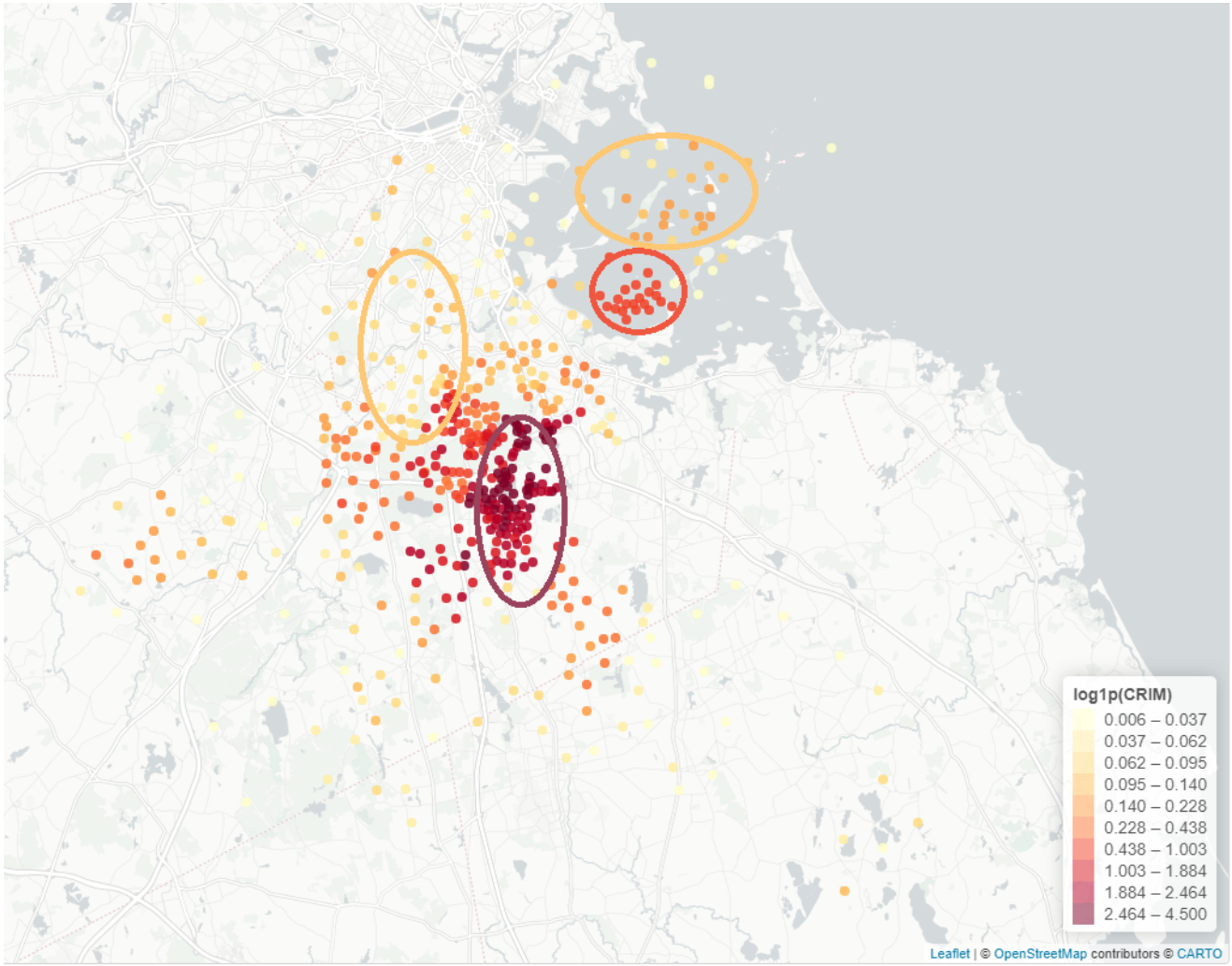

Figure: Visualisation of the log of the variable CRIM

Mathematical background (1/2)

- If we zoom out a bit and evaluate the situation, finding a window Z where the means inside and outside are different is the same as identifying Z as a cluster.
- Now than we defined the logic of the test we need to determine which test to use to compare the means inside and outside the window Z.
- We have two main properties regarding the log likelihood under H_0 & H_1

Mathematical background (2/2)

- log-likelihood of $(X_1, ---, X_n)$:
  $ln(L_0) \underset{H_0}{=} -n * ln(\sqrt{2\pi}) - n * ln(\sigma) - \sum_i \frac{(x_i-\mu)^2}{2\sigma^2}$
  - $\hat{\mu}_{ML} = \frac{\sum_i x_i}{n}$
  - $\hat{\sigma}^2_{ML} = \sum_i \frac{(x_i-\mu)^2}{n}$

- log-likelihood of $(X_1, ---, X_n)$:
  $ln(L_Z) \underset{H_1}{=} -n * ln(\sqrt{2\pi}) - n * ln(\sigma_Z) - \frac{n}{2}$
  - $\hat{\mu}_{Z_{ML}} = \frac{\sum_Z x_i}{n}$
  - $\hat{\sigma}^2_{Z_{ML}} = \frac{1}{n}[\sum_Z (x_i^2 - x_Z\mu_Z + n_Z\mu_Z^2) + (\sum_{\not{Z}} x_i^2 - 2(X - x_Z)\lambda_Z + (n - n_Z)\lambda_Z^2)]$

    with:
    - $x_Z = \sum_Z x_i$ & $\mu_Z = \frac{x_Z}{n_Z}$ & $\lambda_Z = \frac{X - x_Z}{n - n_Z}$
    - We recognize λ_Z as the mean outside the window

Statistic computation

- Now that we have defined the log-likelihood under both H_0 and H_1, we can formulate the maximization problem:

$$\begin{aligned}\text{Scan}_\mu &= \max_Z \Big(\ln(L_Z) - \ln(L_0) \Big) \\ &= \max_Z \left(n \ln(\sigma) + \sum_i \frac{(x_i - \mu^2)}{2\sigma^2} - \frac{n}{2} - n \ln(\sqrt{\sigma_Z^2}) \right)\end{aligned}$$

- This statistic is called a scan statistic and so the test is called a **scan test**. The only term that depends on Z is the last one : $-n * ln(\sqrt{\sigma_Z^2})$ so we can deduce a property: the window Z that satisfies the maximization problem, and so, will be the solution of our maximization problem is also the one that minimizes its variance.

Generating the null distribution: Permutation (Bootstrap)

- The theoretical distribution of the scan statistic $Scan_\mu$ is unknown.
- We approximate it by generating artificial datasets under the null hypothesis.

- **Key idea:** destroy spatial structure while preserving the data distribution.
  - Keep locations (coordinates) fixed
  - Randomly permute the observed values x_i across locations

- For each permutation r:
  - Obtain a new dataset $x_i^{(r)}$
  - Compute the corresponding statistic $Scan_\mu^{(r)}$

- Repeat this process B times to generate a reference distribution.

Monte Carlo test for the p-value

- We compare the observed statistic to the simulated ones.
- Steps:
  - 1) Compute the observed statistic $Scan_\mu$
  - 2) Generate B permuted statistics $Scan_\mu^{(r)}$
  - 3) Compare the observed value to the simulated distribution
- The Monte Carlo p-value is:

$$p_B = \frac{1}{B+1} \sum_{r=1}^{B} \mathbf{1}_{\{Scan_\mu^{(r)} \geq Scan_\mu\}}$$

- Interpretation:
  - If $p_B < \alpha$, we reject the null hypothesis
  - The observed clustering is unlikely to be random

Example (1/3)

- To compute the scan test, we relied on **rsatscan**, an R package that interfaces with the external software **SaTScan**.
- This solution was the most convenient and efficient, because SaTScan performs the scan statistic computation, while R handles data preparation and post-estimation analysis.
- More precisely, the code:
  - prepares the case, coordinate, and parameter files required by SaTScan,
  - launches the scan procedure from R,
  - and imports the output back into R to analyse the detected spatial clusters.

Example (2/3)

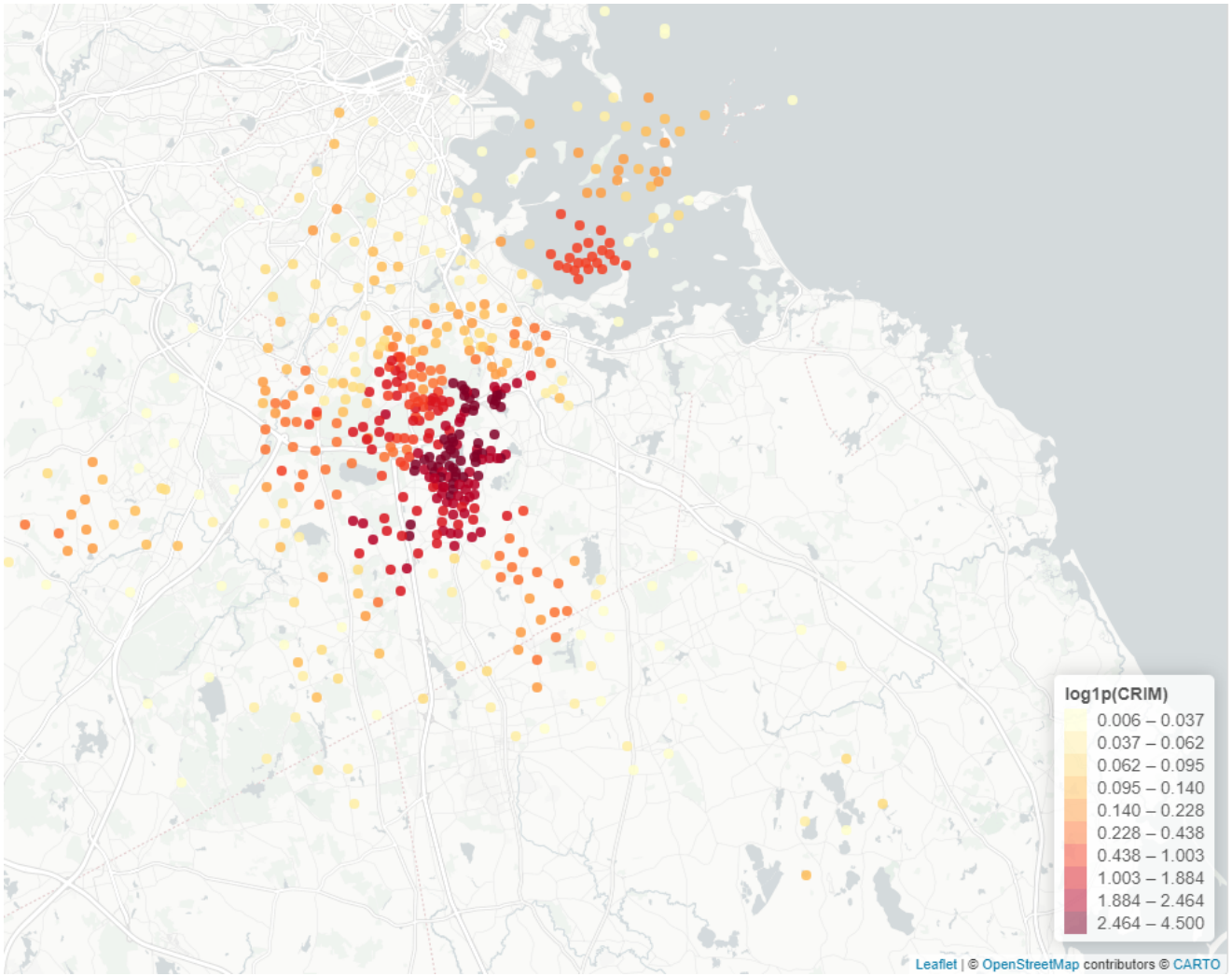

Figure: Geo visualisation of CRIM

Example (3/3)

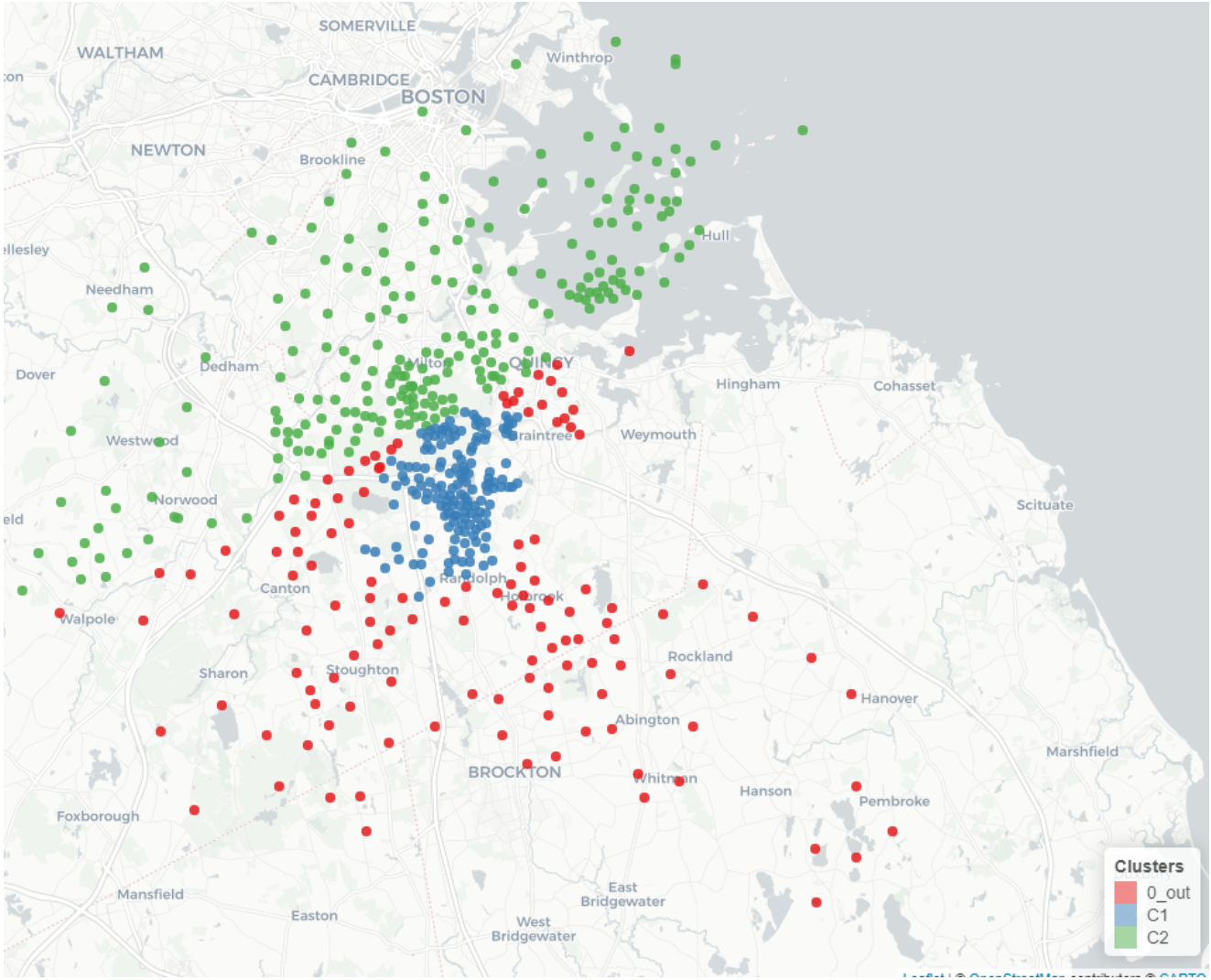

Figure: Geo visualisation of clusters of CRIM

Most likely cluster (MLC)

- Let us suppose we reject $H_0 \Leftrightarrow$ we identify a cluster. We name this cluster the **Most Likely Cluster**.
- Once this cluster is found, it can still be useful to search for other clusters, they are the **secondary clusters**. The issue is that, if we follow the same algorithm as the one to find the MLC one, we will find a cluster that overlaps with the MLC. We need to add a condition
  - 1) We select new windows Z' that doesn't overlap with the window Z of the MLC
  - 2) Then we follow the same algorithm and maximize the $Scan_{\mu}$
  - 3) We compute the p_B-value by comparing the likelihood ratio of such secondary clusters with the MLC
  - 4) Then we can rank the possible secondary clusters according to their likelihood ratio

Section 3

Special application to model residuals

Residuals case: general idea

- In the univariate case, the scan test compares the mean value inside the window Z with the mean value outside Z.
- In this case, this comparison cannot be extended directly in a simple way when several explanatory variables are involved.
- The idea is therefore to first estimate a model including all relevant covariates, and then apply the scan test to the **residuals** rather than to the raw variable itself.
- Intuitively, the residuals represent the unexplained part of the outcome after controlling for the observed variables.

Residuals case: implementation

- In practice, we first estimate a regression model with the explanatory variables of interest.
- We then compute the residuals and use them as the input for the scan procedure.
- The scan test therefore detects areas where the residuals are significantly higher or lower inside the window Z than outside.
- In our code, this is exactly the logic followed: estimate the model, extract the residuals, and run the scan test on these residuals through **rsatscan / SaTScan**.

Residuals case (Example)

- In the paper, the authors apply the scan test to a **hedonic housing price model** estimated on geo-referenced housing data for **Madrid**.
- Idea : start from a standard econometric specification explaining housing prices with structural and neighbourhood characteristics, and then analyse the **residuals** of the model.
- They use the scan test to detect **local spatial clusters** in these residuals, that is, areas where the model systematically over- or under-predicts prices.
- They then compare this information with more standard tools of spatial econometrics, such as **Moran's I** and spatial model diagnostics, to improve the specification of the housing price equation.

Residuals case (Example)

- The clusters detected by the scan test provide **additional location information**: they reveal zones where prices share a common local effect that is not captured by the observed covariates.
- Economically, this may correspond to **unobserved neighbourhood effects**, such as local amenities, accessibility, prestige, environmental quality, or other place-specific characteristics.
- Including cluster or location indicators in the model therefore helps to capture this **omitted spatial heterogeneity**.
- In the paper, the interest of this strategy is precisely to use the scan test as a **specification tool**: detected clusters help explain residual spatial structure and improve the econometric model.

Application

- We chose three different models and will use them to identify clusters and compare the results across models:
  - An OLS model
  - A SEM model

OLS Model (1/8)

- We used an OLS with all the variables presented at the beginning of this presentation:

$$CMEDV_i = \beta_0 + \beta_1 CRIM_i + \beta_2 ZN_i + \beta_3 INDUS_i + \beta_4 AGE_i + \beta_5 TAX_i + \beta_6 PTRATIO_i + \epsilon_i$$

OLS Model (2/8)

Variable	Coefficient	Std. Error	p-value
(Intercept)	54.1160	3.3549	< 0.0001***
CRIM	-0.1714	0.0464	0.0002***
ZN	0.0169	0.0186	0.3624
INDUS	-0.2778	0.0807	0.0006***
AGE	-0.0196	0.0163	0.2308
TAX	-0.0017	0.0033	0.5994
PTRATIO	-1.4098	0.1773	< 0.0001***

Table: OLS estimation results

Statistic	Value
Multiple R^2	0.3844
Adjusted R^2	0.3770
F-statistic	51.9300
Model p-value	< 0.0001***

Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

OLS Model (3/8)

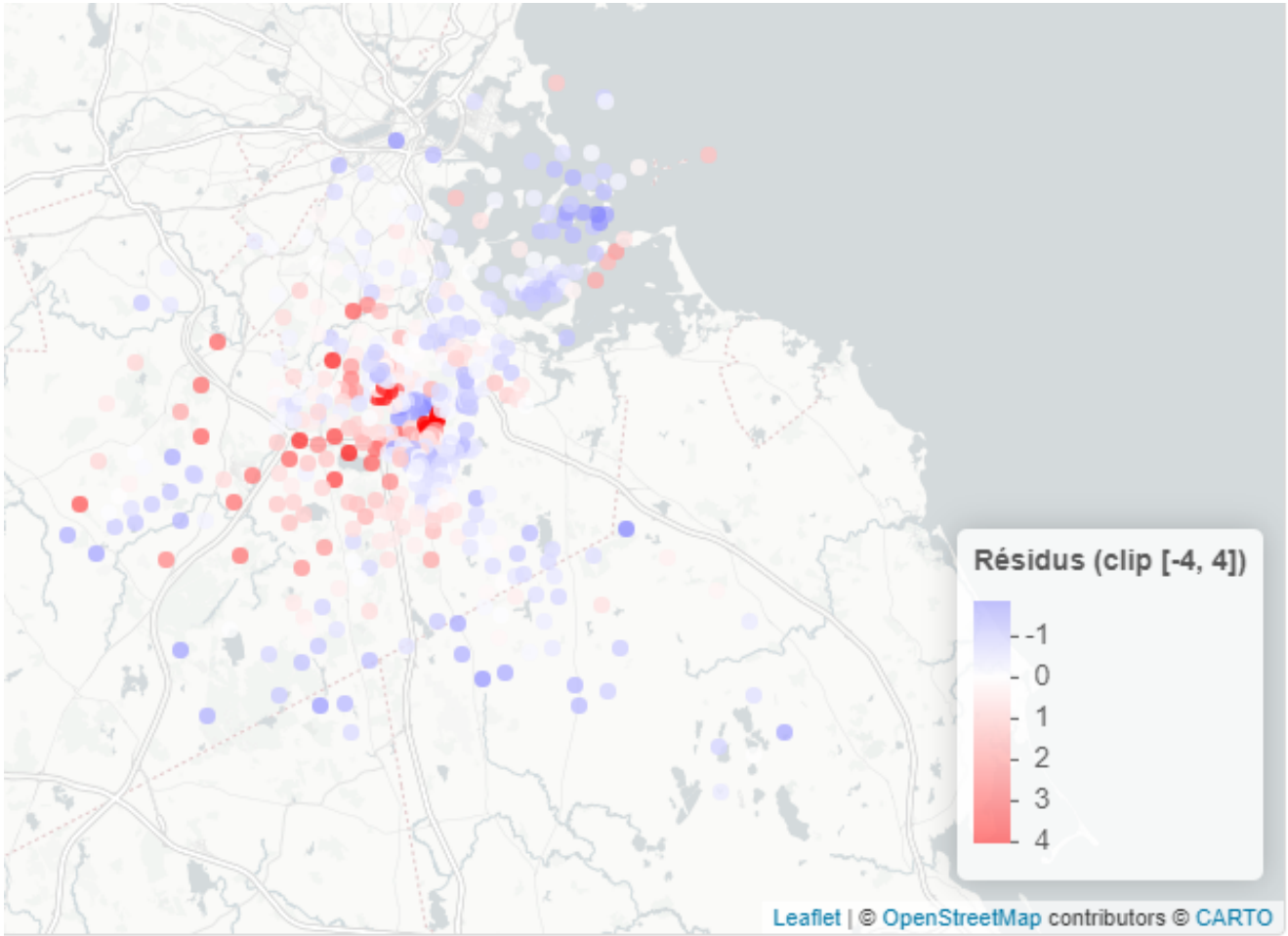

Figure: Geo visualisation of residuals from the OLS model

OLS Model (4/8)

Cluster	Observed	Mean in	Mean out	Std. Dev.
1	5	4.6669	-0.0466	0.8844
2	122	0.7014	-0.2228	0.9184
3	200	-0.3987	0.2606	0.9465

Table: Main scan test cluster characteristics

OLS Model (5/8)

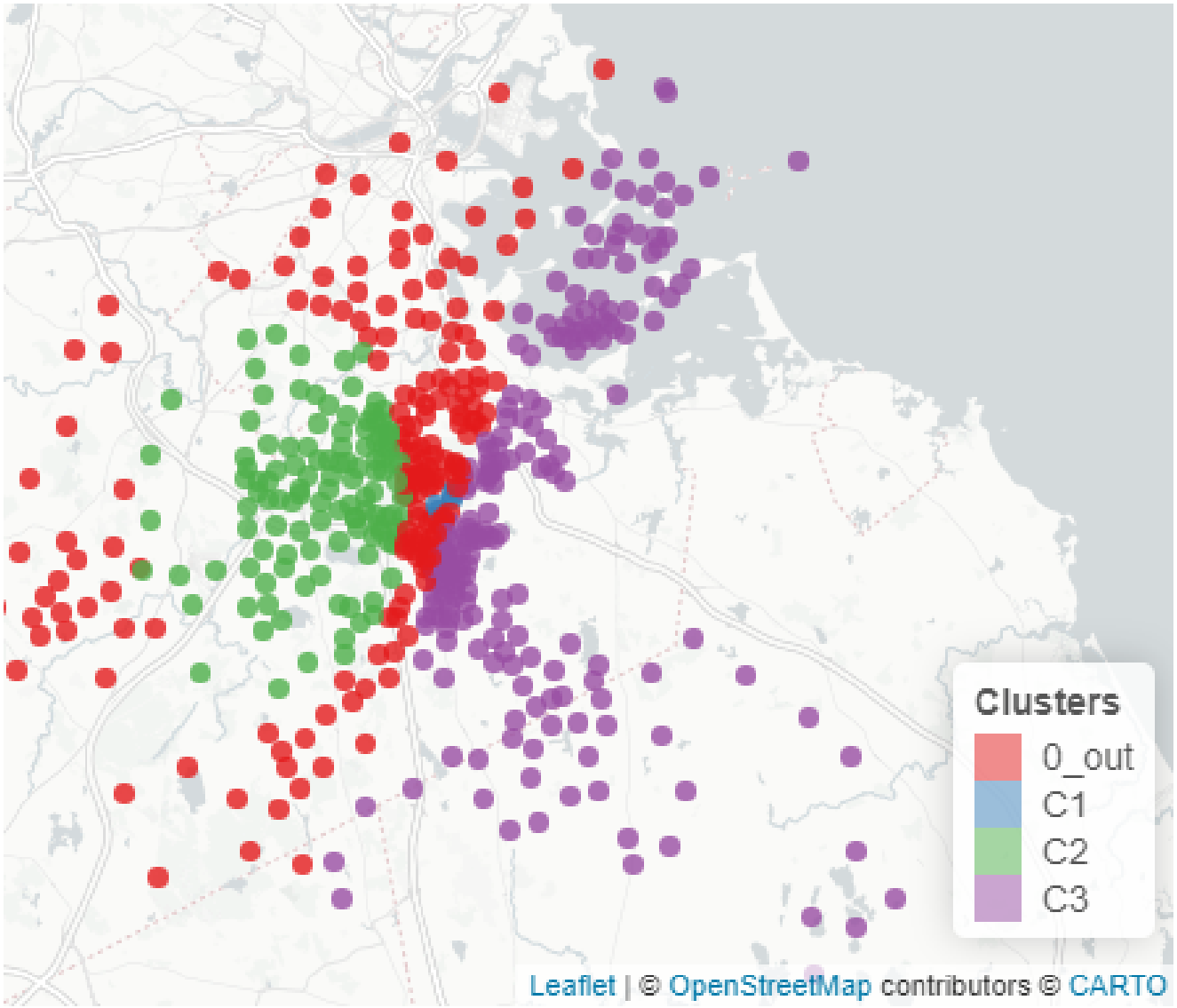

Figure: Geo visualisation of residuals from the OLS model

OLS Model (6/8)

- We used an OLS with all the variables presented at the beginning of this presentation:

$$CMEDV_i = \beta_0 + \beta_1 CRIM_i + \beta_2 ZN_i + \beta_3 INDUS_i + \beta_4 AGE_i + \beta_5 TAX_i + \beta_6 PTRATIO_i + \beta_7 CLUSTER_ID_i + \epsilon_i$$

OLS Model (7/8)

Variable	Coefficient	Std. Error	p-value
(Intercept)	41.4602	2.7298	< 0.0001***
CRIM	-0.1054	0.0350	0.0027**
ZN	0.0375	0.0141	0.0082**
INDUS	-0.2471	0.0634	0.0001***
AGE	-0.0388	0.0125	0.0020**
TAX	-0.0048	0.0025	0.0562*
PTRATIO	-0.7024	0.1434	< 0.0001***
cluster_id1	34.8692	2.5043	< 0.0001***
cluster_id2	7.4676	0.6907	< 0.0001***
cluster_id3	-2.0082	0.5987	0.0009***

Table: OLS estimation results with cluster indicators

Statistic	Value
Multiple R^2	0.6573
Adjusted R^2	0.6511
F-statistic	105.7000
Model p-value	< 0.0001***

OLS Model (8/8)

Model	Moran's I	Std. deviate	p-value
Initial OLS residuals (*resid*0)	0.4945	19.025	< 0.0001
Residuals with cluster effects (*resid_m*1)	0.3450	13.263	< 0.0001

Table: Comparison of Moran's I tests

- Both tests reveal strong and significant positive spatial autocorrelation in the residuals.
- However, after introducing the cluster effects, Moran's I decreases from **0.4945** to **0.3450**.
- This shows that the detected clusters capture an important part of the spatial structure left unexplained by the initial OLS model.
- The spatial dependence is reduced, but not fully removed, which suggests that some residual spatial pattern still remains.

Conclusion on the OLS model

- The initial OLS model explains part of housing prices, but its residuals still display strong and significant spatial autocorrelation.
- The scan test detects several clusters in these residuals, showing that the model leaves important local spatial structure unexplained.
- Once cluster indicators are added to the regression, the fit improves substantially, and Moran's I decreases.
- However, spatial autocorrelation remains significant, which means that the OLS specification is still not sufficient to fully account for spatial dependence.

SEM Model(1/5)

- Finally we used a SEM model:

$$CMEDV_i = \beta_0 + \beta_1 CRIM_i + \beta_2 ZN_i + \beta_3 INDUS_i + \beta_4 AGE_i + \beta_5 TAX_i + \beta_6 PTRATIO_i + u_i,$$

- with $u_i = \lambda \sum_j w_{ij} u_j + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$

SEM Model (2/5)

Variable	Coefficient	Std. Error	p-value
(Intercept)	46.2891	4.0084	< 0.0001***
CRIM	-0.1073	0.0372	0.0039***
ZN	0.0376	0.0176	0.0321**
INDUS	-0.2924	0.0989	0.0031***
AGE	-0.0747	0.0171	< 0.0001***
TAX	-0.0122	0.0039	0.0017***
PTRATIO	-0.4685	0.1996	0.0189**

Table: SEM estimation results

Statistic	Value
λ	0.8594***
LR test	266.8900***
Wald statistic	1170.4000***
Residual variance (σ^2)	26.7770
Residual std. error (σ)	5.1747
AIC	3184.5000
AIC (OLS)	3449.3000

SEM Model (3/5)

Observed	Mean in	Mean out	Variance	Std. Dev.
5	21.4247	-0.2138	22.2400	4.7159
5	15.9822	-0.1595	24.2758	4.9271
6	11.3151	-0.1358	25.2907	5.0290
7	10.2671	-0.1440	25.3484	5.0347

Table: Main cluster characteristics from the scan test

SEM Model(3/5)

- We now add the information of clusters for each observation:

$$CMEDV_i = \beta_0 + \beta_1 CRIM_i + \beta_2 ZN_i + \beta_3 INDUS_i + \beta_4 AGE_i + \beta_5 TAX_i + \beta_6 PTRATIO_i + \beta_7 LOCATION_ID_i + u_i,$$

- with $u_i = \lambda \sum_j w_{ij} u_j + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$

SEM model (4/5)

Variable	Coefficient	Std. Error	p-value
(Intercept)	40.5912	2.9346	< 0.0001***
CRIM	-0.0558	0.0287	0.0517*
ZN	0.0299	0.0134	0.0252**
INDUS	-0.3117	0.0739	< 0.0001***
AGE	-0.0730	0.0129	< 0.0001***
TAX	-0.0109	0.0029	0.0002***
PTRATIO	-0.3030	0.1516	0.0457**
cluster_sem1	31.2973	2.3453	< 0.0001***
cluster_sem2	24.5847	2.4560	< 0.0001***
cluster_sem3	12.6280	1.8536	< 0.0001***
cluster_sem4	14.3093	1.8357	< 0.0001***

Table: SEM estimation results with cluster indicators

Statistic	Value
λ	0.8035***
LR test	218.0800***
Wald statistic	622.1300***
Log-likelihood	-1442.0780
Residual variance (σ^2)	15.7310
Residual std. error (σ)	3.9662
AIC	2910.2000
AIC (LM)	3126.2000
Observations	506

Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

SEM Model (5/5)

Model	Moran's I	Std. deviate	p-value
SEM residuals	0.0237	1.2449	0.1066
SEM + clusters residuals	-0.0150	-0.6319	0.7363

Table: Comparison of Moran's I tests

- In both specifications, Moran's I is not statistically significant, which indicates that the SEM already removes the spatial autocorrelation present in the OLS residuals; adding the cluster effects slightly reduces it further, but without a major change in statistical conclusion.

Conclusion on the SEM model

- The SEM specification performs better than the OLS model because it directly accounts for spatial dependence through the error structure.
- This is confirmed by the strong significance of λ, the lower AIC, and the absence of significant residual spatial autocorrelation.
- The scan test still identifies local clusters, and including them further improves the fit of the model.
- Overall, the SEM appears to be the most appropriate specification in our application, while the scan test remains useful to detect and interpret local spatial heterogeneity.

OLS vs. SEM: comparison

- OLS improves our understanding of housing prices, but leaves strong residual spatial autocorrelation.
- SEM provides a better fit by directly modelling spatial dependence in the error term.
- In the OLS model, cluster indicators reduce Moran's I, but do not eliminate spatial dependence.
- In the SEM model, residual spatial autocorrelation is no longer significant, and clusters mainly capture additional local heterogeneity.
- Therefore, SEM is preferable in our application, while the scan test complements both approaches.

Section 4

Conclusion

General conclusion

- The scan test is a useful tool to detect **local spatial clusters** and to complement standard tools of spatial econometrics.
- In our application, it highlights areas where the model leaves a specific local structure unexplained, and therefore provides additional information on spatial heterogeneity.
- The comparison between OLS and SEM shows that modelling spatial dependence directly is essential: the SEM clearly performs better in removing residual spatial autocorrelation.
- At the same time, the scan test remains valuable because it helps identify and interpret localized effects that global indicators such as Moran's I do not describe precisely.
- Overall, the scan test should be seen less as a substitute for spatial econometric models than as a **complementary specification and diagnostic tool**.

Limits

- Even if the results we displayed seemed encouraging, there is still limitations worth motioning
  - 1) The method only works for static data (time dimension not included).
  - 2) To compute the test statistic, we require finite variance, which is a strong assumption (homoskedasticity).
  - 3) In the residuals case, the algorithm works on residuals of a model; if the model is not appropriate, the resulting identified clusters may be unstable.

Kulldorff. A scan statistic for continuous data based on the normal probability mode. *International Journal of Health Geographics*, 2009.

Fernando A. López. Exploring scan methods to test spatial structure with an application to housing prices in madrid. 2013.